

CONTRACTGUARD AI

Local-first contract risk detection for OpenAPI REST APIs and JavaScript/TypeScript consumers.

Field	Details
Course	Web Programming & Mobile Applications Development
Institution / Department	Khulna University, CSE Discipline
Supervisor	Prof. Dr. Kazi Mashudul Alam
Team Members	Md. Farid Hossen Rehad , ID : 220222 Md. Sarafat Karim , ID: 220216
Repository	https://github.com/Schr0Smi1ey/ContractGuard-AI.git

1. Project Scope

Purpose

ContractGuard AI is a planned local-first developer tool that detects probable contract drift and integration risks between REST APIs described using OpenAPI and JavaScript/TypeScript applications that consume those APIs. It compares the available API contract with selected consumer-side usage and presents evidence for developer review.

In Scope

- OpenAPI 3.x REST API specification analysis.
- JavaScript/TypeScript consumer-code analysis.
- Detection of probable API contract mismatches involving endpoints, HTTP methods, parameters, and request or response schemas.
- GitHub repository connection or analysis where required by the MVP.
- Presentation of detected findings with supporting evidence for developer review.
- Local Ollama assistance for advisory explanations of detected findings.
- Local-first execution suitable for a university project demonstration.

Out of Scope

- GraphQL, gRPC, SOAP, and WebSocket or event-stream contract analysis.
- Consumer-code analysis for Python, Java, C#, Go, or other non-JavaScript/TypeScript languages.
- Enterprise-scale repository governance, autonomous code merging, or production CI/CD enforcement.
- Cloud AI dependency, full security vulnerability scanning, and production-scale infrastructure.

2. Project Objectives

1. **Analyze OpenAPI contracts** — Parse OpenAPI REST API specifications and extract endpoints, methods, parameters, request structures, and response structures required for contract comparison.
2. **Analyze JavaScript/TypeScript API consumers** — Identify common REST API usages in JavaScript/TypeScript code, including fetch and axios patterns where feasible.
3. **Detect contract inconsistencies** — Compare provider contracts with consumer-side API usage and identify probable mismatches such as missing endpoints, incorrect methods, incompatible parameters, or schema changes.
4. **Provide understandable findings** — Present detected contract risks with sufficient evidence, severity or context, and clear explanations so developers can review the issue.
5. **Use local AI assistance responsibly** — Use a local Ollama model where useful to improve the readability of requirement text and detected-risk explanations while keeping deterministic analysis and human review authoritative.
6. **Produce a demonstrable MVP** — Build a focused local prototype capable of demonstrating selected API contract-change scenarios within the university project scope.

3. Team Roles and Responsibilities

Md. Sarafat Karim — Requirements, Frontend & Project Coordination

Primary responsibilities:

- Requirements analysis and SRS refinement.
- Project scope and objective validation.
- Frontend/UI implementation and frontend-backend integration.
- Coordination of project milestones and documentation consistency.
- Usability testing, integration testing, and peer review of teammate work.

Md. Farid Hossen Rehad — Backend, Contract Analysis & Integration

Primary responsibilities:

- Backend/API implementation.
- OpenAPI parsing and processing.
- JavaScript/TypeScript consumer analysis and contract comparison.
- Contract mismatch-detection logic, GitHub integration, and local Ollama/AI integration.
- Backend and analyzer testing and peer review of teammate work.

Shared Responsibilities

The following responsibilities are shared by both members and do not constitute a separate project role:

- Technical design decisions.
- Git collaboration.
- Testing and debugging.
- Final integration.
- Code review.
- Final project demonstration.
- Final documentation review.